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Measurement of the cognitive load in mobile situations using wearable devices plays important roles in smart human-computer interactions, physical health monitoring, mental health monitoring, and so forth in daily life. Among wearable devices, wristbands are similar to traditional watches and suitable for most scenarios, such as classrooms and offices; thus, they are widely accepted and becoming increasingly popular. However, most wristbands are only used for step counting and heart rate monitoring. In this paper, we propose a cognitive load monitoring method based on wristband heart rate sensors. The sensors used to detect heart rate are generally photoplethysmogram sensors, but the current major manufacturers extract only the heart rate parameter of the photoplethysmogram. We make full use of the waveform of the wristband photoplethysmogram signal to measure real-time cognitive load, and establish a real-time cognitive load measurement system, which expands the application scope of the wristband photoplethysmogram sensor and shows great potential for measuring cognitive load in daily life. In experimental validation using an n-back task, our waveform-based CNN model achieved approximately 20% higher accuracy in cognitive load classification versus traditional feature-based methods (e.g., LR, SVM, and GNB), reaching an average of 80.11%. The system successfully classifies the cognitive load and demonstrates the practical viability of the proposed approach.

CCS Concepts: • **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → **Cognitive science**; **Perception**.

Additional Key Words and Phrases: photoplethysmogram, cognitive load, wearable

1 Introduction

The concept of cognitive load was first proposed in the education field [1, 2]. The main theory is cognitive load theory (CLT), and the key tenet of CLT is that a higher cognitive load will lower learning ability. In recent years, cognitive load has been widely used in game design [3, 4], user interface design [5], workload measurement [6, 7], automotive design [8, 9], programming learning [10, 11], and so forth. The measurement of cognitive load in the mobile situations will greatly expand the scope of applications in real life.

There is a long history of measuring and monitoring the physical and mental status of individuals. In recent decades, different measuring and monitoring devices have been developed and validated. These devices, such

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as electroencephalogram (EEG) equipment, electrocardiogram (ECG) equipment, blood pressure measuring equipment, and saliva measuring equipment, are mainly operated by professionals in clinical or laboratory settings. These devices require the help of professionals to prepare, correctly wear, control operations, and draw conclusions. Therefore, the measurement and monitoring of human factors are very complex and face difficulty in daily life situations.

However, with the progress of technology, such as machine learning, the gradual development and utilization of micro-sensors and monitoring technology integrated into wearable devices, as well as the large amount of data stored on cloud platforms, the complex operation of human factor measurement has been simplified [12]. In particular, the rise of wearable devices has pushed human factor measurement into practical applications. One of the most obvious features of wearable devices is their mobility [13], which greatly increases the scope of application, integrating the original laboratory equipment into equipment in daily life. In addition, the new wearable technology is relatively affordable and is being rapidly developed and improved, providing possibilities for different workplace or overload environments, and helping to monitor and form interventions in specific application scenarios. At present, the trend of wearable devices is continuously escalating. There are various wearable devices, such as the Polar V800 heart rate monitoring device, Suunto Memory Belt, wearable patch ECG [14], Emotion Board, AutoSense [15], smartwatches [16], and fitness trackers [17]. Because of their convenience, wristband devices are becoming increasingly popular with users, providing efficient interactions for quick access to information and response. In addition to the factors of technological development, another important reason is that in recent years, people have paid increasing attention to a healthier mental and physical lifestyle; thus, the desire to wear is increasing.

Earlier, intelligent wristbands mainly used motion monitoring functions, to analyze calorie consumption, location, and heart rate monitoring. With the rapid development of data processing, wristbands have increasingly more functions. There are an increasing number of studies on consumer products in academia, such as those measuring psychological stress, cognitive load [18], and physical condition. Three types of bracelets on the market, the Apple Watch, Microsoft Band, and Fitbit Surge, have been surveyed in the literature, revealing that wearable sensors can be used to detect poor physiological patterns in the resting state and to detect some threats to mental or physical health [19]. Most wristband devices contain built-in sensors for detecting heart rate signals, and the sensors used to detect heart rate are typically photoplethysmography (PPG) sensors [20, 21], so it is possible to measure cognitive load on the basis of widely worn wristband PPG sensors. However, because these sensors are incorporated within wearable devices, their performance is limited, mainly in terms of the sampling rate and signal intensity. The position of the wristband acquisition signal is proximal to major blood vessels, but the blood pulse of the wrist is relatively weak, which causes a weak PPG signal [22]. Moreover, given the different vascular structures, the morphology is quite different from that of the finger. Traditionally, the PPG signal is acquired from the finger, and until recently, most research on PPG has been based on finger clips. In this way, the PPG signal of the wristband will lose its original standard form, and it is difficult to extract features via traditional methods. A comparison between the signal of the wristband PPG sensor collected on the front of the wrist and the signal of the finger clip PPG sensor collected on the index finger is shown in Figure 1.

There are many problems caused by waveform changes. First, the characteristics are no longer of physiological significance. Many previous studies on PPG physiology have been based on fingertip PPG. Second, the wristband waveform mostly exhibits poor diastolic waves, some of which have higher diastolic amplitudes than the systolic wave or lack diastolic waves. In this case, the features based on diastolic waves, which leads to many effective features, cannot be extracted [23]. A low sampling rate leads to a large feature error, such as a 20 Hz sampling rate; the error of the RR parameters is 0.01s, the feature based on the time dimension is affected, and fewer features can be extracted from the second or third derivatives [24]. In general, feature-based cognitive load measurement methods, such as those in this paper [25], are not feasible for wristband data analysis.

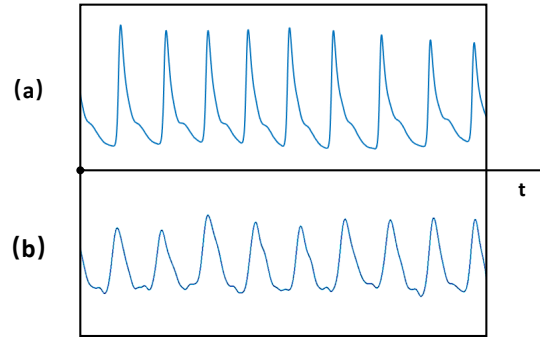


Fig. 1. Comparisons between finger-clip PPG sensor data and wristband PPG sensor data. (a) Finger clip PPG sensor data acquisition location: index finger; (b) Wristband PPG sensor data (acquisition position: front of wrist)

In this paper, we propose a cognitive load monitoring method based on wristband PPG, which does not extract the characteristics of wristband PPG and directly uses the whole waveform and neural network processing methods to make full use of the information of PPG signals, to achieve high accuracy in the classification of cognitive load. The main contributions of this paper are as follows:

- Cognitive load identification method based on waveforms and neural networks. Most of the PPG signals acquired from wristbands are not sufficient to extract all kinds of features. The whole waveform is input into the multilayer neural network to classify the cognitive load.
- Real-time cognitive load identification system based on the Empatica E4 wristband. The cognitive load model is trained based on the data collected in the experiment, and a real-time cognitive load assessment system is developed to evaluate the cognitive load in real time.

2 Related Work

Wearable devices are developing rapidly with the advancement of hardware integration technology. In addition to wristbands, various wearable devices, such as gloves, glasses, helmets, and finger rings, have been developed. However, other devices, with the exception of wristbands, are less popular and are not sufficiently comfortable to wear in daily life. Therefore, this section reviews only the related work on cognitive load measurement based on wristband devices. At present, wristband devices include electrodermal activity (EDA) sensors, PPG sensors, temperature sensors and acceleration sensors. The measurement of cognitive load based on wristband devices is based mainly on these three sensors.

2.1 Research on Cognitive Load Measurement Using EDA Sensors

Some studies have used EDA sensors on wristbands, such as [26] EDA sensors of the E4 wristband with a sampling rate of 4Hz, to analyze the difference in the cognitive load of processing information when screens and paper are used. The results show that the principal components of EDA reading on screen are 73.38% greater than those of reading on paper. Setz et al. [27] used EDA sensor data from an emotion board to differentiate between stress and cognitive load. The main problem associated with using EDA sensors is the accuracy of long-term use. The value

of EDAs is strongly affected by the ambient temperature, and long-term exposure to sweat strongly affects the results; thus, EDA does not meet the needs of daily use and long-term wear.

2.2 Cognitive Load Measurement Using PPG Sensor

Heart rate information from a PPG in a wristband is typically used to categorize cognitive load. Romine et al. [28] used it to monitor cognitive load in a learning environment, and modeling revealed that heart rate increased with increasing mental focus. When the heart rate increased by one standard deviation, the probability of a focused state increased by up to 1.4 times [29] when the cognitive load was evaluated solely on the basis of heart rate information from the PPG, with 66% accuracy in classifying the cognitive load as high, medium, or low. Given that some heart rate characteristics require 3-5 minutes to obtain a value, it is not easy to achieve real-time results on the basis of heart rate information.

2.3 Cognitive load Measurement Using Multiple Sensors

Some studies use all the sensors on the wristband to assess cognitive load. For example, the literature [18] uses GSR, temperature, heart rate and heart rate variability to classify cognitive load. However, the accuracy of the two classifications using the 10-fold cross validation method and excluding one subject's data is only approximately 53%, which is close to 50%, indicating random classification accuracy. Even for individual classifications, the accuracy is between 66% and 86%. The features adopted in this research are the average GSR, temperature, mean and standard deviation of heart rate in each time period, the RMSSD of the continuous difference in the RR interval, and the percentage of RR intervals (pNN50 and pNN20) that differ by more than 50 ms and 20 ms, respectively, from the previous RR interval. The low accuracy of this method may be due to the unreliability of some features in the extraction process; for example, heart activity may increase because of other factors in the subjects [30]. Furthermore, not every wearable device necessarily contains these sensing data together, making the method unfavorable for widespread dissemination.

3 Cognitive Load Measurement Based on Waveform

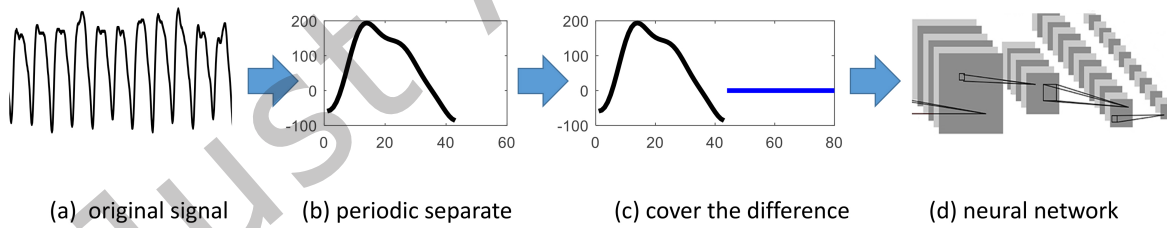


Fig. 2. Waveform-based cognitive load detection. (a) original wristband PPG data; (b) periodic separation; (c) each waveform is supplemented to 80; (d) input neural network

Previous image processing methods are based on image extraction features, such as corners, edges, and SIFT features, but the recognition accuracy needs to be improved. Since the emergence of the neural network, images can be directly input into the neural network. Different layers of the neural network can automatically extract different feature layers, such as the pixel layer, feature layer, and region layer. This greatly improves

the recognition rate. At present, the purpose of PPG involves extracting various features; however, the features cannot fully describe the waveform, and the features reflect only one aspect. Therefore, image processing can be used as a reference to input the whole waveform into the neural network for cognitive load classification, which can make full use of the information of the whole waveform and avoid the problem of incomplete consideration of artificial feature extraction.

We propose waveform-based cognitive load detection. The entire process is shown in Figure 2. First, we obtain the initial PPG signal from the wristband. The initial PPG signal may fluctuate depending on the subject's wearing position, the gap between the wrists, and the subject's motion artifacts. The frequency ranges of the PPG signals and motion artifacts typically fall within 0.5–4 Hz and 0.1–10 Hz, respectively [31]. The spectral overlap between these bands makes it difficult to separate artifact components from the useful signal. To address this, the FFT (fast fourier transform) algorithm is employed in this study to convert the signal from the time domain to the frequency domain. For a discrete sequence $x[n]$ of length N , Equation (1) is applied to decompose the signal into a superposition of multiple sine and cosine waves.

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-j2\pi \cdot \frac{kn}{N}} \quad (1)$$

To attenuate frequency components outside the dominant range of the PPG signal, all frequency components in $X[k]$ below 0.5 Hz and above 4 Hz were set to zero, thereby removing out-of-band noise. However, since motion artifacts spectrally overlap with the useful signal, and considering that PPG signal characteristics are manifested primarily in the time domain, an inverse FFT was applied. This involved performing a complex conjugate operation on the frequency-domain signal to revert it to the time domain for further filtering. Next, we chose an FIR filter (finite impulse response) to refine the PPG signals. FIR is a popular digital filter that is capable of filtering without changing the phase information of the signal, which means that the filtered signal does not have distortion. This feature makes it particularly suitable for phase-sensitive applications such as ECG (electrocardiogram) and speech signals. Compared with the IIR (infinite impulse response), the system function of the FIR is poleless, causing no filter destabilization because of numerical errors, and remains stable at higher orders, allowing the restoration of more accurate PPG signals with more details. FIR can be optimized with different window functions; for PPG signals, we use the Kaiser window, which is based on the adjustable parameter β to flexibly control the width of the main flap and the attenuation of the side flaps, and to effectively suppress the leakage of the spectrum.

$$w(n) = \frac{I_0 \left(\beta \sqrt{1 - \left(\frac{2n}{N-1} - 1 \right)^2} \right)}{I_0(\beta)} \quad (2)$$

where $I_0(x)$ is the zeroth-order modified Bessel function of the first kind:

$$I_0(x) = \sum_{m=0}^{\infty} \frac{(x/2)^{2m}}{(m!)^2} \quad (3)$$

Equation (2) defines the Kaiser window function, and Equation (3) represents the modified Bessel function of zeroth order.

Afterward, the PPG signal is subsequently cut into a single heartbeat waveform via periodic separation. With respect to the separation method, this study describes a novel signal decomposition approach based on SVD (singular value decomposition) [32], which has been successfully applied to extract fetal electrocardiogram signals from composite maternal electrocardiogram signals. The core of the algorithm involves applying SVD to a windowed matrix of the signal and observing the distribution of singular values to identify periodic boundaries.

In this study, the PPG signals were first reconstructed by constructing a Hankel matrix using a sliding window approach.

$$H = \begin{bmatrix} x_{i+1} & x_{i+2} & \dots & x_{i+M} \\ x_{i+2} & x_{i+3} & \dots & x_{i+M+1} \\ \vdots & \vdots & & \vdots \\ x_{i+M} & x_{i+M+1} & \dots & x_{i+2M-1} \end{bmatrix} \quad (4)$$

In Equation (4), x_i represents the parameter value of the PPG signal at time i , whereas M denotes the window size designed to cover one complete PPG cycle.

For each window M , the SVD decomposition yields $\Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_n)$, where the singular values σ are arranged in descending order. The SVR (singular value ratio), an SVD-based algorithm, can be employed to detect signal periodicity. A local minimum in the SVR indicates a signal transition or the onset of a new cycle—that is, the boundary point of an individual heartbeat waveform.

$$SVR(t) = \frac{\sigma_1}{\sum_{i=1}^r \sigma_i} \quad (5)$$

To ensure the physiological validity of the individual waveforms, the filtered and segmented signals were subjected to a fivefold validity check. This includes verification of the cycle duration (0.8–1.6 s), peak location, presence of dual peaks, trough location, and monotonicity of the rising phase. Valid individual heartbeat waveforms may vary in length. To be input into the neural network, they must be the same size. The sampling rate of the wristband is 64 Hz, and the normal heartbeat of a human is between 50 and 100 beats per minute. Therefore, the length of a single heartbeat waveform is between 38.4 and 76.8; thus, we set the length of all single waveforms to 80 by adding a zero-offset to the end of each waveform. All the PPG waveforms of a single heartbeat are subsequently input into the neural network.

Artificial neural networks have been widely used in the image, text, and sequence fields. Using artificial neural network recognition technology, the PPG waveform of each heartbeat is input into the neural network as a feature. To recognize the cognitive load, the structure of the model needs to extract the features in the image automatically. The current PPG signal is weak and lacks the original standardized shape. In this paper, we choose the classic and excellent convolutional neural network (CNN) for image recognition and classification [33]. Convolution is the critical step in the CNN, which can adjust the area of feature extraction and output the feature map according to the convolution kernel. Working with a pooling layer allows downsampling of the image, reducing the amount of data and preserving valuable features. This mechanism is very beneficial for processing physiological images with feature correlation.

4 Experiment

To validate the methodology of this paper, we conducted a classic memorization-based cognitive load experiment, the n -back task, to stimulate different levels of cognitive load, with the data collected and analyzed.

4.1 Subjects

We recruited 20 subjects aged between 18 and 22 years, including 14 women and 6 men. All the subjects had similar educational backgrounds with no previous similar experiments to reduce the effects of individual ability or background on performance. The experiments were approved by the local ethics committee and performed in accordance with the guidelines of the Declaration of Helsinki. Each subject was paid 20 dollars for participation.

4.2 Apparatus

In the experiments, all the subjects were required to wear two PPG sensors, a finger-clip PPG sensor and a wristband PPG sensor, as shown in Figure 3. The finger-clip PPG sensor used was the HKG-07 Infrared Digital Pulse Sensor (Hefei Huake Institute of Electronic Technology, China), with a sampling rate of 200Hz, the same equipment as in some previous works [3, 34, 25] on cognitive load. Thus, the finger-clip PPG sensor can be used as a reference gold standard.

We used an Empatica E4 wristband (Empatica Inc., United States) to obtain a wearable PPG signal. The E4 wristband can provide the original PPG signal with a sampling rate of 64Hz, and the wristband development document is clear, provides timely technical support, and is friendly to developers. In a previous study to test the pressure detection performance of a wearable sensor system, the E4 wristband was compared with an AD Instruments Powerlab ECG acquisition device used in a stationary laboratory ([35]). In this study, 14 HRV characteristics in the time and frequency domains of seven subjects were used, and the pressure discrimination ability of the characteristics with respect to different acquisition devices was compared. The E4 wristband was found to have significant deficiencies in detecting heart rate intervals; but was able to successfully estimate time domain characteristics such as the mean heart rate and standard deviation of the heart rate. The E4 wristband achieved good discrimination ability against mental stress, with an identification efficiency of 87.5%. The E4 wristband can provide a real-time heart rate interval, but this interval cannot be given for every heartbeat; thus, direct application of this interval signal is problematic. In this paper, we used the original signal from the PPG, so there is no case of interval loss that affects the results. The Empatica E4 wristband PPG sensor contains two pairs of light sources and light receivers. The two light sources are different: one is a green light source, which contains the main information of the heartbeat, and the other is a red light source, which contains the information of movement. By combining the signals of two light receivers, a PPG sensor can remove more motion artifacts [36] than can a single light source PPG sensor. In daily life, especially in mobile environments, this property confers more advantages.

During the experiment, the PPG sensor with a finger-clip was worn on the left index finger, whereas the wristband was worn on the left wrist.



Fig. 3. Two PPG sensors used in the experiment

4.3 Stimulations of Cognitive Load

The *N-back* is one of the most widely used and established tasks for stimulating cognitive load [37]. This task is easy to implement and does not require complex experimental equipment. In this experiment, the subjects needed to watch consecutive letters appearing on the screen (shown in Figure 4), and for each letter, they were required to determine whether it was a target or a non-target by pressing a key, with the left key representing the

target and the right key representing the non-target, and the experiment was divided into three stages: *0-back*, *1-back*, and *2-back*.

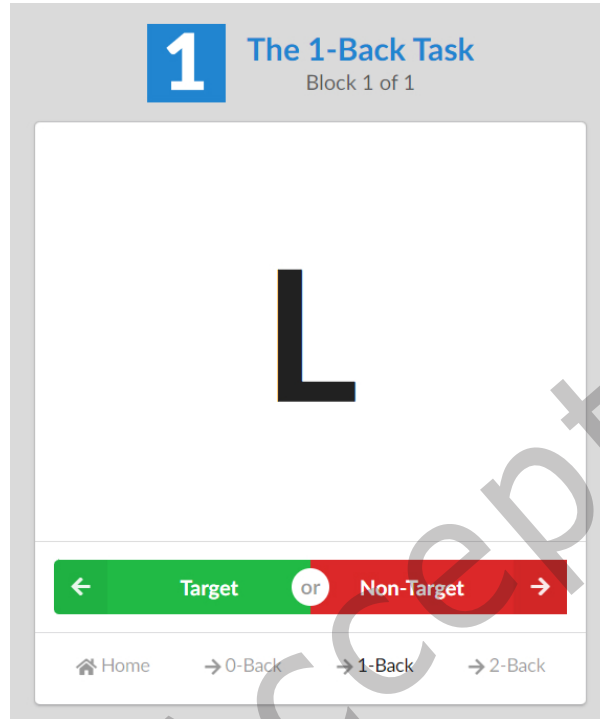


Fig. 4. *n*-back task experiment interface

- In the *0-back* experiment, the condition was that a letter appeared to be a target if it was the fixed stimulus—the letter “X”.
- In the *1-back* experiment, the condition was that a letter was a target if it appeared as the same letter as the previous one, otherwise, it was a non-target.
- In the *2-back* experiment, the condition was that a letter was a target if it appeared the same as the first two occurrences, otherwise, it was a non-target.

To help the subjects stay focused and inspired, screen changes in red and green and different types of sound feedback were provided when they gave wrong and correct answers. In addition, points were awarded. The subjects received 1 point for a correct answer and no points for a wrong answer.

4.4 Protocol

The goal of the study was to elicit three different states (low cognitive load, medium cognitive load, and high cognitive load) in the subjects. Upon arrival at the study location, the subjects were equipped with the HKG-07C+infrared digital pulse sensor and Empatica E4 wristband.

The complete procedure lasted about 30 minutes—10 minutes for the tutorial exercise and 20 minutes for the formal experiment. The tutorial exercise consisted of the experimenter assisting the subjects in wearing the measuring instruments and explaining the rules and procedures of the experiment. The experiments were

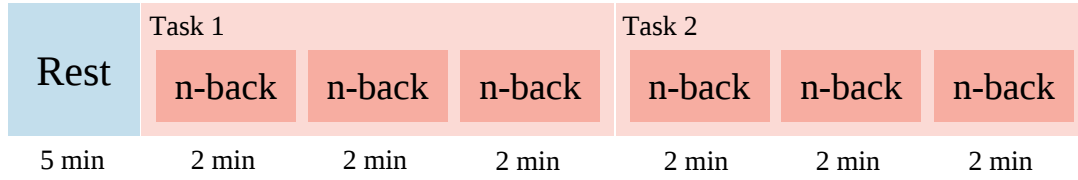


Fig. 5. Protocol of experiment

grouped into two stages (task1 and task2), and the three conditions (*0-back*, *1-back*, and *2-back*) were presented in 2-minute time blocks in each stage. The order of the three time blocks in each stage was randomized for participants, with different time blocks between the end of the first stage and the beginning of the second stage.

At the end of each stage, the number of subjects, scores, number of correct answers and number of incorrect answers were stored in a text file. The entire experimental process is shown in Figure 5.

4.5 Self-evaluation

To validate the study protocol, the subjects were required to perform a self-evaluation at the end of the experiment, as shown in Figure 5. We used the RSME scale to investigate subjective mental effort. The scale ranges from 0 to 150, with larger values representing increased workload. After each *n-back* task, the subjects assessed the value of their subjective mental effort for the current mission. The values of the scale can be taken as a subjective report of how the participants felt during each task.

4.6 Classification Model Design

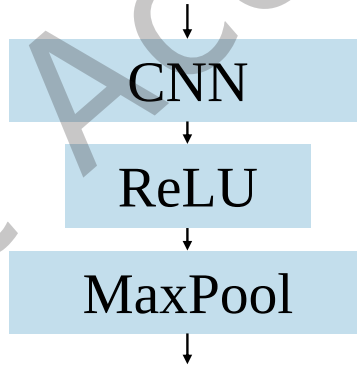


Fig. 6. Core structure of the model

As described in Section 4.2 of this paper, the Empatica E4 wristband has two PPG optical sensors. Empatica uses a set of security algorithms to combine two signals, and users can obtain only the fitted signal. Therefore, the characteristics of finger-clip PPG, as shown in Figure 1, lose their physiological importance when applied to wristband PPG. Therefore, we use the cognitive load classification method based on the PPG waveform in this paper.

After standardizing the length of the sample sequence (as shown in Section 3 of this paper), we used normalization to eliminate the magnitude and keep the distributions of the dimensions close to each other to ensure

successful model convergence. In this paper, we set up the model as a three-classification prediction model to recognize the user's cognitive load at three intensities: 0-back, 1-back, and 2-back, respectively. Figure 6 shows the core structure of the model, where we use the CNN convolutional power to capture the temporal characteristics of the signals and MaxPooling to retain the most prominent features and reduce computational effort. The model consists of four sets of core blocks, each of which chooses ReLU as the activation function to accelerate training, effectively solves the vanishing gradient problem, and provides sparse activation when processing data to improve computational efficiency. The CNN and Pooling layer in each block use the same cores. In the first two blocks, the size of the kernel is 3, and the size of the kernel in the last two blocks is 2. The Dropout layer is used between the blocks to prevent overfitting of the model. The model concludes with a softmax activation function that transforms the output of the network into a categorical probability distribution, where each value represents the probability of the corresponding category, and the probabilities sum to 1. The model uses Adam as the optimizer and sets the learning rate to 0.001, with a loss function of CategoricalCrossentropy.

To validate the model in this paper, we introduced Long Short-Term Memory (LSTM) for comparison. Changes in PPG signals can be considered a serialization problem, and LSTM is the current typical model structure that can preserve historical information. LSTM performs well in pattern recognition tasks based on stream input, such as speech recognition, language modeling, handwriting recognition, etc. The comparison with LSTM provides a clear representation of the CNN model in this experiment.

5 Results and Discussion

In the experiment, 18 subjects completed all the tasks. Therefore, we obtained 18 sets of raw data and used them all for analysis.

5.1 Performance and Self-reports

The task and self-evaluation results verified that the experiment was able to control for the subjects' cognitive load. In addition, they were employed to confirm whether participants devoted the expected amount of effort.

The accuracies of the three tasks are shown in Figure 7. The accuracy and score exhibit the same trend. The accuracy for the 0-back task was 0.989 ± 0.016 , that for 1-back task was 0.935 ± 0.045 , and that for 2-back task was 0.906 ± 0.068 . The increasing difficulty of the tasks resulted in progressively lower accuracy with statistically significant differences of $F = 13.053$ and $p = 0.000$. The results prove that the three tasks truly differed in terms of difficulty. In other words, the task was correctly set.

Table 1 shows the results of the three self-reported measures (see subsection 4.5), respectively.

Table 1. Mean, standard deviation and p value of the three self-reported measures

Items	0-back	1-back	2-back	p
RSME	10.306 ± 13.146	44.944 ± 29.420	74.083 ± 37.458	0.000

In the 0-back, 1-back, and 2-back conditions, the mean scores on the RSME were 10.306, 44.944, and 74.083, respectively, with $F = 43.842$, $p = 0.000$, reconfirming the significant differences across conditions and the correctness of the experimental setup. Tough tasks caused subjects to exert more effort, even with participants who chose to RSME of 150 after the 2-back task. Therefore, more difficult tasks impose a greater cognitive load on subjects, whereas simple tasks impose a lower cognitive load on participants.

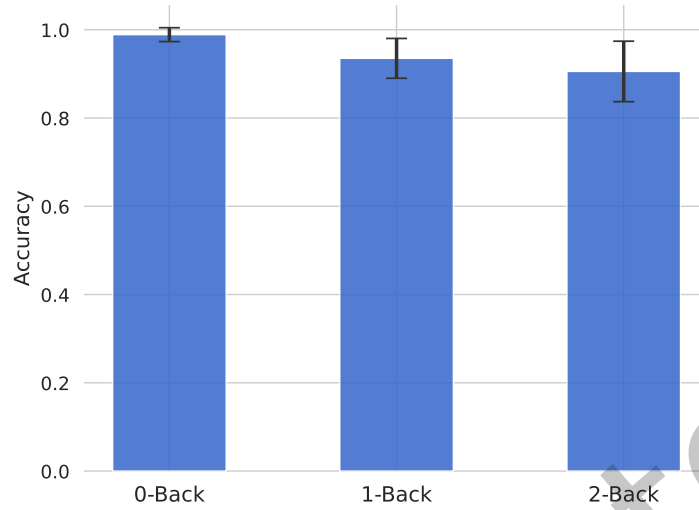


Fig. 7. The Performance of the subjects in Experiment

5.2 Statistical Results

First, the results of the finger-clip PPG are analyzed. Several commonly used indicators of PPG are listed, including Stress-Induced Vascular Response Index (sVRI), the systolic peak amplitude (PA), the heartbeat interval (RR) and the area under the curve (AUC) of the PPG waveform. As shown in Figure 8, sVRI is an effective and reliable index for evaluating cognitive load [38] and $sVRI = A2/A1$. PA is an index of vasoconstriction in peripheral blood circulation. Heart rate interval (RR) is R-R interval and is often used as heart rate. AUC is the geometric index of PPG signal. All of these parameters were obtained by a PPG sensor worn on the subjects' left index finger. The baseline value is the average value of the data in the first and second remaining stages. The results are shown in Table 2.

Table 2. Statistical Means of Finger-clip PPG at Different Stages

feature	0-back	1-back	2-back
sVRI	0.8852	0.8984	0.8957
PA	117.7423	111.6374	107.1580
RR	0.7606	0.7643	0.7202
AUC	65.8834	67.2553	63.2381

Moreover, the significance of each parameter in different stages is calculated in Table 3. As seen from the results, the sVRI and RR features were both non-significant in the two neighboring tasks, and they were highly significant only in the 0-back vs. 2-back, suggesting that the differences between the tasks were not large enough. The PA and AUC features were significant in all three comparisons, indicating that the task-stimulated levels of cognitive load can be disentangled.

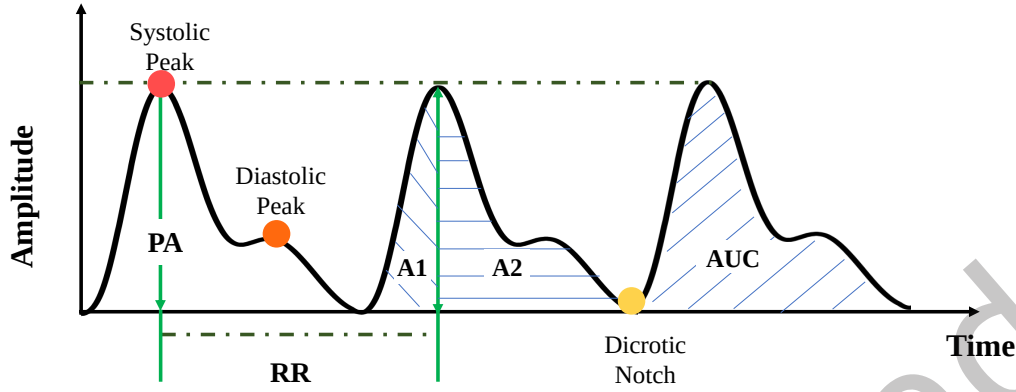


Fig. 8. Features of finger-clip PPG, including the systolic peak (PA), wave range amplitudes (A1 and A2), heart rate interval (RR) and area under the curve (AUC). [3]

Table 3. Significance of finger-clip PPG at different stages, expressed as (F , p).

feature	0-back vs. 1-back	0-back vs. 2-back	1-back vs. 2-back
sVRI	(19.932, 0.000)	(14.023, 0.000)	(1.025, 0.311)
PA	(28.602, 0.000)	(81.3, 0.000)	(13.532, 0.000)
RR	(1.137, 0.286)	(143.702, 0.000)	(169.39, 0.000)
AUC	(17.333, 0.000)	(71.756, 0.000)	(163.253, 0.000)

5.3 Classification Results

We classify the individual independence as "leave one subject out". The individuals in the test set and the training set are independent; that is, the data in the training set do not appear in the test set. We retained the data of one subject in the test set and used all the data of the remaining subjects in the training model. Therefore, for the data from 18 subjects, the process was repeated 18 times. The results are shown in Figure 9. The minimum recognition accuracy is 74.61%, and the average recognition accuracy is 92.34%.

The accuracy of nine of the subjects was greater than 80%, and that of only one was greater than 90%, indicating that the recognition level of the model differs from person to person, but remains at overall superior level. The right Figure shows the average accuracy using different structures; the CNN model has an average accuracy of 80.11%, which is significantly higher than that of the LSTM and SVM models (57.76% and 50.99%, respectively), verifying the superiority of our model in terms of accuracy. Meanwhile, the LSTM model learns poorly during training, with severely slow convergence, and the model is not stable enough for different subjects, with an accuracy difference larger than 40% (99.7% and 50.22%). These findings indicate that the CNN model can simultaneously achieve strong generalization ability.

5.4 Comparison with Multi-feature Method

In previous papers, the cognitive load classification method based on PPG morphological features has also achieved good results; in particular, the accuracy of classification and recognition based on 46 features can reach 100% in some cases [25]. Therefore, for comparison, we also calculated the wristband PPG data on the basis of multiple features. Compared with those of the transmitted finger-clip PPG sensor data, the waveforms of wristband PPG data are more severely distorted and lose the standard PPG waveform shape, with some samples having no dicrotic wave. Moreover, the waveform contrast is shown in Figure 1. Therefore, on this basis [25], all 46 features cannot be extracted. The features associated with the dicrotic peak would be impossible to compute owing to the possible lack of dicrotic waves in the wristband data. Ultimately, 22 features can be extracted from the wristband PPG data over a wide range, as shown in Table 4. The specific parameter meanings can be referred from [25].

With these 22 features, the classification accuracy is calculated by using seven classifiers, namely, LR, SVM, GNB, DT, RF, AdaBoost, and GB, which are commonly used in machine learning classification problems. The results are shown in Table 5. From the table, it can be determined that the classification accuracy based on features is approximately 60%, while the classification accuracy based on waveforms and neural networks is 80.11%. The waveform-based method proposed in this paper improves the accuracy by approximately 20%. There are two main reasons for the improvement in accuracy: on the one hand, the limited number of effective features extracted does not fully contain the effective information of PPG waveform for morphological changes when the cognitive load changes; on the other hand, the method based on a neural network automatically extracts features in the back network, which greatly improves the utilization of signal morphological features.

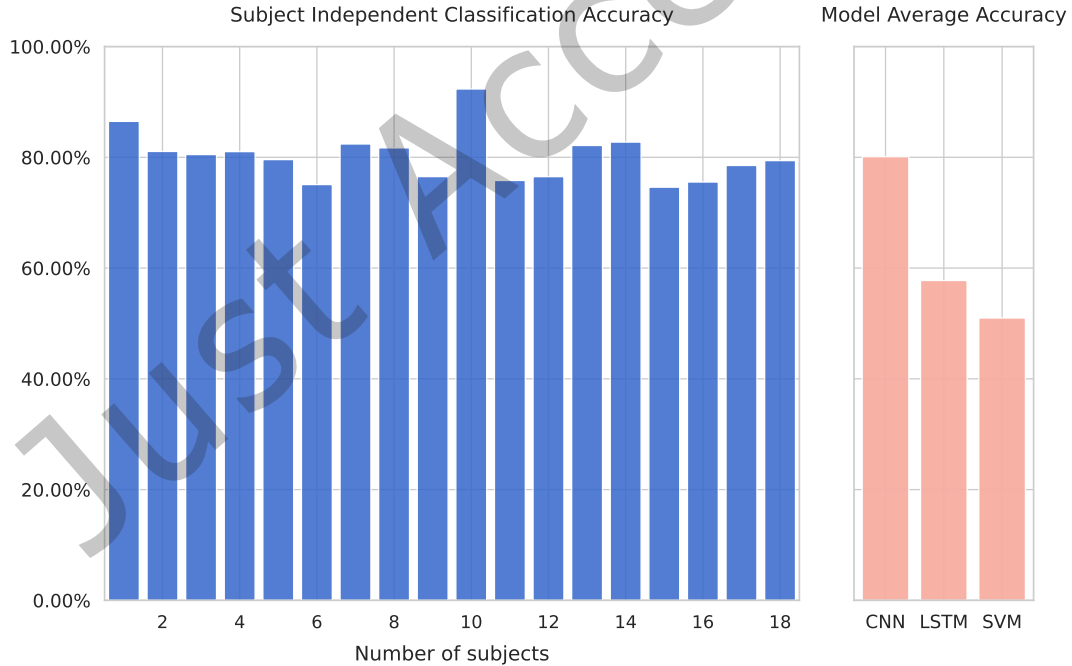


Fig. 9. Model accuracy for different subjects and structures. The left panel shows the results of the CNN model in different subjects, and the right panel shows the results of the comparison of the average detection accuracy between different models.

Table 4. Definition of Wristband PPG-based features (PPG-22)

#	Feature	Description	#	Feature	Description
1	x	Systolic peak	2	t_{pi}	Pulse interval
3	w	Time between half systolic peak points	4	$(A_2 + A_3)/A_1$	Stress-Induced Vascular Response Index (sVRI)
5	t_1/x	Systolic peak rising slope	6	t_{pi}	
7	t_{a1}		8	t_{b1}	
9	t_{a2}		10	t_{b2}	
11	t_{a1}/t_{pi}		12	t_{b1}/t_{pi}	
13	t_{a2}/t_{pi}		14	t_{b2}/t_{pi}	
15	$(t_{a1} + t_{a2})/t_{pi}$		16	$(t_{b1} + t_{b2})/t_{pi}$	
17	f_{base}	Fundamental component frequency	18	$ s_{base} $	Fundamental component magnitude
19	f_{2nd}	Second harmonic frequency	20	$ s_{2nd} $	Second harmonic magnitude
21	f_{3rd}	Third harmonic frequency	22	$ s_{3rd} $	Third harmonic magnitude

Table 5. Classification accuracy of wristband data based on 22 features

Classifier	LR	SVM	GNB	DT	RF	AdaBoost	GB
Accuracy	61.05%	61.88%	58.83%	64.49%	62.94%	59.76%	60.96%

6 Design and Implementation of Real-time Wristband-based Cognitive Load Assessment System

To demonstrate the practicality of this method in real life, a real-time cognitive load assessment system based on wristband PPG data was designed. The system is composed of a wristband with a PPG sensor and a computer or smartphone. All the devices should have Bluetooth wireless connectivity or Wi-Fi connectivity. The real-time PPG signal is transferred using Bluetooth from the wristband to a computer or a smartphone, which realizes cognitive load recognition based on the PPG signal. To assess the cognitive load level based on the real-time PPG signal, we constructed a neural network model using all the data in the experiment in Section 4.

The specific design of the system is shown in Figure 10. First, real-time PPG data are acquired from the wristband. Typically, the wristband is equipped with a Wi-Fi module or Bluetooth module. The Empatica E4 wristband equipped with a Bluetooth module is used in this system. Therefore, the PPG data are transferred to a Dell computer using Bluetooth module in real-time. All data processing is conducted on a Dell G315.6 with an Intel Core i7 processor, 8GB of memory and a 6G independent graphics card, running Windows 10. The whole system can run smoothly on the computer. Once the computer receives the data, the data processing begins. First, the data are low-pass filtered with a 3rd-order butterworth filter. Second, the data are separated into one heartbeat waveform. Third, each heartbeat waveform is supplemented to 80 points and input into the constructed neural network model. The prediction of the cognitive load by the model is accelerated by the GPU of the computer. Finally, the current cognitive load level is displayed in real time on the interface of the computer, and the cognitive load curve is recorded.

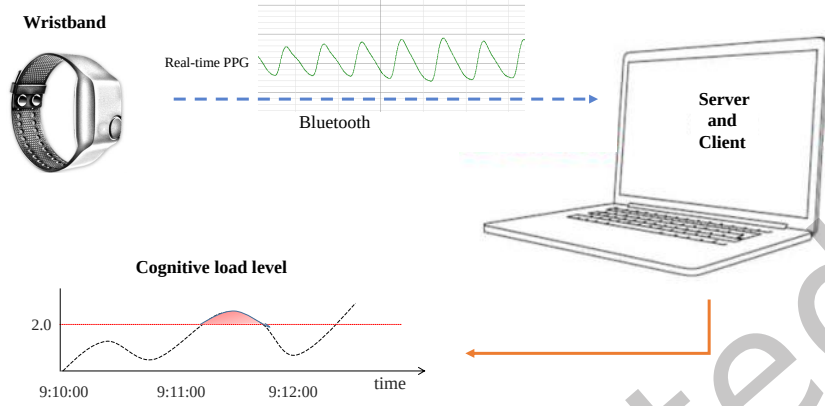


Fig. 10. Real-time Cognitive Load Measurement System Based on Wristband

The GUI of the system is shown in Figure 11. The upper column shows the PPG data of the wristband received in real time, and the lower right column shows the evaluated real-time cognitive load level based on these data. The lower right and middle columns show the operation of the connected wristband and the communication results between the wristband and the computer.

7 Conclusion

In this paper, a real-time cognitive load measurement method for wearable devices using wristband PPG is proposed. The PPG signal collected by wristband equipment is quite different from the signal collected by transmission finger-clip PPG sensor; for example, the rebound wave is not obvious. Therefore, the conventional method based on feature extraction has difficulty obtaining the available performance. In that case, in data processing, the whole PPG waveform is input into the neural network for cognitive load assessment, which is based on the idea of image processing in a neural network. An experiment was implemented to validate the method using the Empatica E4 wristband wherein arithmetic tasks were used to stimulate cognitive load. The experimental results show that the accuracy of the method based on waveforms and neural networks can reach 80.11%, and compared with that of the traditional feature extraction method, the accuracy is improved by approximately 20%. It is confirmed that the waveform-based cognitive load recognition method proposed in this paper can effectively identify different levels of cognitive load. Using wristband PPG data collected in the experiments, a neural network for cognitive load assessment is established, and a real-time cognitive load assessment system is designed and implemented, which shows great potential for measuring the cognitive load in daily life.

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Fig. 11. A screen-shot of the GUI of Real-time Cognitive Load Measurement System Based on Wristband

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